Remote sEnsing based multi-sensor Spatio-tEmporal fusion aided ReserVOIR MAPPing framEwoRk (RESERVOIR MAPPER)

Framework Overview

- **Scope & Objective:** This framework can be used across diverse climatic and geographic regions worldwide to generate dense and temporally continuous reservoir water spread time-series.
- **Multi-Sensor Fusion Strategy:** Integrates high-resolution optical (Sentinel-2) and Synthetic Aperture Radar (Sentinel-1) datasets to overcome cloud cover limitations by replacing cloud-contaminated optical pixels with data from the temporally nearest SAR water scene. * Utilizes spatial fusion techniques where cloud-contaminated optical pixels are replaced by data from the temporally nearest SAR water scene.
- **Platform:** Built entirely on the Google Earth Engine (GEE) platform, leveraging its high-performance cloud computing and extensive data catalog.
- **Details:** For more details regarding the methodology, please refer to the paper "Remote sEnsing based multi-sensor Spatio-tEmporal fusion aided ReserVOIR MAPPing framEwoRk (RESERVOIR MAPPER)" by Shekhar and Ramsankaran (2026) (DOI: 10.1016/j.rsase.2026.102054).

Implementation Instructions

- **Setup:** Paste the "Reservoir_Mapper" code into the Google Earth Engine Code Editor.
- **Geometry Input:** Input the lake geometry using a buffered boundary to ensure the area covers the maximum potential water spread and seasonal fluctuations.
  - **Option A (Manual Digitization):** Use the GEE drawing tools to create a new polygon layer around the reservoir directly on the map.
  - **Option B (Shapefile Import):** Go to the GEE 'Assets' tab, click 'New' then 'Shape files', upload your zipped shapefile component files (.shp, .shx, .dbf, .prj), and import it into your script.
- **Execution:** Run the code within the GEE platform.
- **Data Export:** To access the water spread time-series for longer time periods (e.g., several years), Export the CSV output to your Google Drive.
  - **Steps:** Locate the 'Tasks' tab, click the 'Run' button next to the task, configure your preferred Drive folder path in the popup window, and click 'Run' again to begin the background export.
  - **Note:** This method bypasses the "Capacity Exceeded" or "Computation Timed Out" warnings during processing on GEE.

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